

An Unfinished Symphony: Completing the Playbook for Reading Education



Why researchers, teachers, and writers now argue America's reading crisis is a **professionalization problem — not a knowledge one.**

For many, the Science of Reading movement has entered its 'trough of despair'. Not only are nationwide scores not up; they are, in fact, down. Finger-pointing has begun. "Too much phonics". "Not enough whole books." "Time wasted on phonemic awareness." "MTSS doesn't work."

The uncomfortable truth is that structured literacy suffers from the same confusion as "balanced literacy". It remains — in its fine-grained elements — under-defined.¹

The problem shifts: How to make good practice consistent?

For decades, reading reform centered on getting evidence-aligned instruction into more than a minority of classrooms.

By 2023, Mark Seidenberg was pressing a different question: now that demand for the science of reading had grown, where was the expertise and infrastructure needed to turn a large, changing research base into dependable practice? Teachers should not have to become cognitive scientists; the science has to be built into the systems around them.

By mid-2026, calls for system integration solutions had appeared from a variety of experts. A panel with Doug Carnine, Sarah Oberle, Mike Schmoker, and Robert Pondiscio offers a video synopsis of these.¹¹

In Congress, legislation for an NRP 2.0 has passed out of the Appropriations Committee. If it becomes law, will this solve the problem at hand?

At the scale of millions of teachers and thousands of intermediaries, messages drift as they propagate. Would NRP 2.0 adequately address this?

40% of U.S. fourth-graders read below NAEP Basic in 2024.

Black fourth-graders face massive, systemic hurdles in reading.

From visibility to durability

2023 · SEIDENBERG^{2,3} The science has won attention, but research expertise is thin throughout the education system. Translation and professional infrastructure are missing.

2024-25 · JONES^{4,6} A living profession needs "garbage collection"; other professions have solutions that can help cut instructional waste, reduce opportunity costs, help kids read.

2024-26 · CALLS FOR NRP 2.0^{5,7,8} In the wake of unsustainable and unscaled improvement, calls grow for a new National Reading Panel to synthesize research done since NRP I.

2025-26 · CARNINE/LYON^{9,10,11,12,15} frame the problem as professionalization: shared knowledge, common language, aligned preparation, competence-based licensure, accountability.

2026 · TIPTON/PATTON-TERRY¹³ Beyond the science of reading, we need an "engineering of reading."

THE BEGINNINGS OF PROFESSIONAL INFRASTRUCTURE

The pieces steadily come online

Reading education has begun to acquire much-needed tools and structures.

The Reading League: quality standards for foundational curricula¹⁴

Its Curriculum Evaluation Guidelines and Navigation Reports help schools make better choices.

Reading Rockets + Reading Universe^{16,17}

Reading Rockets has been refurbished and updated; Reading Universe adds a newer, practical, video-rich body of guidance.

State-built professional learning: training becomes infrastructure^{18,19}

States are increasingly building grade-, role-, and context-specific training, administrator pathways, and refresher modules.

UFLI brings a free, quality phonics program, with training.

Scope and sequence, routines, assessments, videos, fidelity tools, and implementation supports package research.

State-Coordinated Coaching Networks²⁰

While 26 states have adopted some form of coaching policies, 8 have the type of coordinated field force that brought Mississippi its gains.

Learning Science Practices | ResearchEd

Led by our friends in the UK, teachers are tying day to day teaching to learning science well beyond structured literacy.

Teacher-prep standards and audits: Fixing the Ed Schools^{21,22}

NCTQ's systematic reviews of teacher-preparation programs — combined with state program reviews — audit whether education schools actually teach structured literacy.

Knowledge Matters: supporting knowledge-building²³

The Knowledge Matters Campaign focuses on the critical role of knowledge-building in early literacy, and on approaches to do so.

Johns Hopkins ELA Knowledge Map²⁴

With the ELA Knowledge Map, reviewers can explore the domains, sequencing, coherence, and gaps in a curriculum's content knowledge.

The EdReports reckoning^{25,26}

The exposure of serious weaknesses in the Ed Reports definition of "evidence-based" has led it to release a "2.0 Review Criteria", and to users being far more circumspect about its ratings.

Podcasts and specialist media: solid learning²⁷

A strong ecosystem of podcast series and specialist videos now lets those with the inclination learn on their own schedule.

Social-Media Dispersal and Discussion

The Facebook Group Science of Reading-What I Should Have Learned in College now has 250,000 members. An extremely well-informed X community daily discusses the hard cases of Structured Literacy. Substack gives longer-form ideas immediate feedback.

R&D and Systems for older readers, too^{28,29}

From more structured fifth-grade word study to systems for severe high school remediation; from individually-authored programs to high end programs like Reading Reimagined, all K12 literacy is getting love.

Building Consistent Nationwide Teaching Capability

No one should doubt the remarkable success represented in the above components. Few, indeed, would have had reason to expect us to have come so far, so fast.

Still Missing

Yet there remains missing an integrating framework. A unifying structure that ties together the various components of the system.

The Gaps — A Case Study

Is Reading Universe a suitable knowledge base? It would appear so: it explains phonemic awareness, decoding, fluency, vocabulary, morphology, and many classroom routines quite well.

Yet critical components are missing: Knowledge-building is not developed as a major segment. Fine-grained issues such as morphophonemic alternation in complex multisyllabic words receive little treatment. Poor practices like teaching “blends” remain.

Further, research traceability is inconsistent and not built into the architecture. It lacks the maintenance, issue tracking, version control, and feedback processes critical to the profession.

Four Approaches

NRP 2.0

While some argue for a new National Reading Panel, the truth is that experts have long been monitoring, sharing, and projecting the updated research. A government-sponsored panel of experts may seem a viable next step. Yet there are better ways to achieve what’s needed.^{7,8}

An ‘Engineering of Reading’

It’s a good phrase: an engineering mindset is definitely needed. Today we have more of a writers + publishers mentality driving the system - extremely useful, yet insufficient to the need.

On the other hand, engineering-like documents can only be intermediate steps. Not teacher- or trainer-facing work products. The various engineering professions are all smaller, and generally draw from a more academically elite demographic base than teaching. Where they can suffer more technical jargon, the teaching profession cannot.

A Field Manual System

Here, the focus is on the median teacher (with various ‘extra duties’, a supplemental coaching contract, 3 kids at home, maybe a parent needing care).³⁰

Part natural ‘Field guide’ (describing nature); part field manual (describing a working system and default operating procedures), not just print, but a science-backed collection of the best all-media resources for learning structured literacy.

Field manuals are something much more than a mere book published by one or two authors, and far more than the standard blog post, video, or podcast episode (not that those don’t serve!), and more than a website of independently authored white papers.³¹

They are a snapshot of the institution’s understanding of best practices: grade-by-grade scope and sequence, explicit routines, decision rules for common classroom situations, diagnostic and progress-monitoring tools, and checklists.

More than that, they are created not just to publish information, but to teach it. They are designed for people who are not first and foremost knowledge workers. They are written with the objective of being read by (in Pondiscio’s words) mere mortals.

They not only explain the research base; they use learning science and instructional principles in their own design for teacher/trainer learning.

A Community-Maintained Canonical Source

Teaching, in the US, is a far larger, and more decentralized profession than most. It has no clear institutional center. Thus, while the US military is a similarly-sized profession, and the US Army’s doctrine and manual system provides a solid model for a knowledge center, reading education has no TRADOC equivalent to organize and maintain such a canonical source.

There is, though, a similarly-sized profession, with similarly decentralized leadership. And they have solved this challenge.

For thirty years, communities of open source software developers have collaboratively built not just products, but documentation and all manner of tools to support the open source processes.

In time, a unique culture also developed, to support those people, methods, and tools. Habits of mind actually changed to support the new model.

Reading education can import that culture and process.

Processes. Named maintainers, contribution rules, issue tracking, version control, review cycles, tests, release histories.

Culture. Contribute, don’t complain. Many eyes make bugs shallow. Being specifically wrong beats being right and vague.

The resulting knowledge base is more reliable precisely because many capable people can inspect it, challenge it, and improve it — without any one organization holding all the expertise.

Moving Past Publish & Forget:

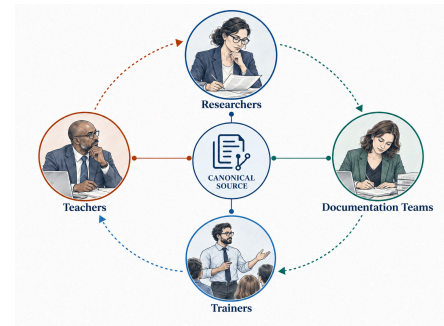
Bringing in Teachers & Trainers; Accelerating the Speed of the Loop

The difference between the two models is not the document. It is whether anything happens after the document ships.

Let's look at some details of practice:

- After K-1 MOY assessments, many teachers are unsure of what to do with the data.

A decision tree is needed. Let's look at making one part of the knowledge base.



The same experts write the same guidance.
THE TWO MODELS ARE IDENTICAL UP TO THIS POINT

PUBLISH AND FORGET

Published as finished: a PDF, an adopted program, a PD day.
THE DOCUMENT IS THE DELIVERABLE

The field hits real defects — gaps in the scope and sequence, texts that don't match the phonics order, guidance that can't be executed.

Teachers fix it privately: personal drives, hallway workarounds, a district file no one else sees.
NO ROUTE BACK TO THE SOURCE

Dead end. Years later a new panel starts over, and everything the field learned is lost.

OPEN SOURCE ETHOS, TOOLS, PROCESSES

One canonical source, numbered and dated. Everyone can say *which* guidance they teach.
VERSION CONTROL

In use nationwide, with every district on a known version — not "the state framework," but v2.1.
RELEASE

KNOWLEDGE COMPOUNDS

A teacher, coach, or district proposes the actual correction — not just a complaint. Maintainers accept or decline in public, with reasons on the record; the fix ships as the next version.
CONTRIBUTION + REVIEW

The defect is filed in the open, with a name and a date on it — not lost in a PD evaluation form or a hallway conversation.
ISSUE TRACKING

And a district can fork it. Local adaptation stays linked to the source instead of drifting away from it, and a good local fix can travel back up. Fidelity and adaptation stop being opposites.

Nothing is thrown away. Every prior version stays citable, and the change from the 2019 guidance to the 2026 guidance is legible line by line — who changed what, when, and on what evidence.

Publishing makes improvement *episodic*: it happens when a panel is convened and funded. Maintenance makes it *cumulative*: the thousands of corrections teachers already make in private become one shared, reviewed, attributed asset that outlives the grant, the superintendent, and the vendor contract.

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